

MS EXCEL

Core Concepts Study Notes

Definition • Syntax • Worked Example — for every function

23 Core Topics Covering Functions, Lookups, Pivot Tables, Power Query, VBA & More

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Important Points for Excel Learners

A short checklist every learner should keep handy — the common errors you will see, the shortcuts that save the most time, and the habits that prevent messy spreadsheets later.

Common Formula Errors & What They Mean

Error	What it means
#N/A	A lookup could not find a match — check spelling, spaces, or data type.
#DIV/0!	The formula is trying to divide by zero or by an empty cell.
#REF!	A cell the formula depended on was deleted.
#VALUE!	The wrong type of value was used — e.g. adding text to a number.
#NAME?	Excel does not recognize the function name — usually a typo.
#####	The column is too narrow to display the value — widen it.

Essential Keyboard Shortcuts

Shortcut	What it does
F2	Edit the active cell directly.
F4	Cycle through Relative / Absolute / Mixed references.
F9	Recalculate all open workbooks.
Ctrl + ;	Insert today's date as a fixed value.
Ctrl + Shift + ;	Insert the current time as a fixed value.
Ctrl + 1	Open the Format Cells dialog.
Ctrl + T	Convert a range into an Excel Table.
Ctrl + E	Run Flash Fill.
Ctrl + Arrow Key	Jump to the last used cell in that direction.
Alt + F11	Open the VBA Editor.

Best Practices Every Excel Learner Should Follow

- Keep one header row and never leave a completely blank row or column inside a data table.
- Avoid merged cells in data tables — they break sorting, filtering, and pivot tables.
- Convert raw ranges into an Excel Table (Ctrl+T) so formulas and formatting extend automatically as rows are added.
- Use Named Ranges for values reused across many formulas (like a tax rate) instead of repeating the same cell reference.
- TRIM and clean imported/copied data before running lookups on it — hidden spaces are the most common cause of lookup errors.
- Freeze the header row (View tab -> Freeze Panes) on any sheet longer than one screen.
- Save a backup copy before recording or running a macro — macros cannot always be undone with Ctrl+Z.
- Use IFERROR around lookups and division formulas that might realistically fail, so reports show a message instead of an error code.
- Document assumptions with Cell Comments/Notes rather than typing notes inside a data cell.
- Save macro-enabled files as .xlsm, and keep a plain .xlsx copy for anyone who does not need the macros.

1. Cell References — Relative, Absolute & Mixed

A cell reference tells a formula which cell to use. There are three types, and choosing the right one matters most when a formula is copied to other cells.

Relative Reference

Definition: Changes automatically to match its new position whenever the formula is copied to another cell.

Syntax: A1 (no dollar sign)

Formula in B2: =A2*2
 Copied to B3: =A3*2
 Copied to B4: =A4*2
 -> each row picks up its own value in column A

Absolute Reference

Definition: Stays completely fixed no matter where the formula is copied — used for one constant value shared by every row, like a tax rate.

Syntax: \$A\$1 (dollar sign before column and row)

Bonus rate stored once in D12 = 10%
 Formula in C2: =B2*\$D\$12 -> 5500
 Copied to C3: =B3*\$D\$12 -> 4500
 D12 never changes in either copy

Mixed Reference

Definition: Locks only the row or only the column, letting the other one move freely when copied.

Syntax: \$A1 locks column A only | A\$1 locks row 1 only

=B2 copied across columns -> column B stays fixed, row changes
 =B\$2 copied down rows -> row 2 stays fixed, column changes

Trainer tip: Press F4 right after selecting a cell reference in the formula bar to instantly cycle through Relative -> Absolute -> Mixed -> Mixed.

2. Flash Fill (Ctrl+E)

Flash Fill

Definition: Studies the pattern in one example you type and instantly repeats that pattern for the rest of the column — no formula required.

How to use:

1. Type the result you want for just the first row next to your data, e.g. type 'Ravi' next to 'Ravi Kumar'.
2. Click the cell below it and go to Data tab -> Flash Fill, or simply press Ctrl+E.
3. Excel detects the pattern from your one example and fills the rest of the column automatically.
4. Scan a few filled rows for exceptions (e.g. names with a middle name) and correct them by hand.

Full Name (raw)	First Name	
Ravi Kumar	Ravi	<- typed manually
Priya Shah	Priya	<- Ctrl+E fills this
Kiran Patil	Kiran	<- Ctrl+E fills this

Trainer tip: Flash Fill pastes plain values, not a formula — if the source data changes later, you need to run it again.

3. Statistical Functions

These are the most frequently used formulas in any Excel report — each one summarizes a list of numbers in a different way.

SUM

Definition: Adds up all the numbers in a range.

Syntax: =SUM(number1, [number2], ...)

Salary column C2:C5 -> 55000, 45000, 78000, 62000
 =SUM(C2:C5) -> 240000

AVERAGE

Definition: Calculates the arithmetic mean (average) of a range of numbers.

Syntax: =AVERAGE(number1, [number2], ...)

=AVERAGE(C2:C5) -> 60000

COUNT

Definition: Counts only the cells that contain numbers, ignoring text and blanks.

Syntax: =COUNT(value1, [value2], ...)

Range has 4 numbers and 1 text entry 'N/A'
 =COUNT(C2:C6) -> 4

COUNTA

Definition: Counts every non-empty cell, whether it holds a number, text, or a date.

Syntax: =COUNTA(value1, [value2], ...)

Same range as above (4 numbers + 1 text)

=COUNTA(C2:C6) -> 5

COUNTBLANK

Definition: Counts only the empty cells in a range.

Syntax: =COUNTBLANK(range)

Range C2:C9 has 2 empty cells

=COUNTBLANK(C2:C9) -> 2

MIN

Definition: Returns the smallest value in a range.

Syntax: =MIN(number1, [number2], ...)

=MIN(C2:C5) -> 45000

MAX

Definition: Returns the largest value in a range.

Syntax: =MAX(number1, [number2], ...)

=MAX(C2:C5) -> 78000

LARGE

Definition: Returns the k-th largest value in a range — the basis of any Top-N ranking.

Syntax: =LARGE(range, k)

=LARGE(C2:C5,2) -> 62000 (2nd highest salary)

SMALL

Definition: Returns the k-th smallest value in a range — the basis of any Bottom-N ranking.

Syntax: =SMALL(range, k)

=SMALL(C2:C5,2) -> 55000 (2nd lowest salary)

Trainer tip: COUNT ignores text entirely; if a column mixes numbers with text like 'N/A', use COUNTA to count all filled cells instead.

4. IFS Function

IFS tests several conditions in a single formula and stops at the first one that is TRUE, avoiding a long chain of nested IFs.

IFS

Definition: Checks multiple conditions in order and returns the result linked to the first TRUE condition.

Syntax: =IFS(condition1, result1, condition2, result2, ..., TRUE, default_result)

```
=IFS(B2<40000,"Junior",B2<90000,"Mid",TRUE,"Senior")
Salary 85000 -> Mid
Salary 120000 -> Senior
```

Trainer tip: Always add a final TRUE, "Default" pair at the end so the formula never returns an error when nothing else matches. IFS works from Excel 2019/365 onward.

5. SUMIF & SUMIFS

Both functions add up numbers that meet a condition — SUMIF handles one condition, SUMIFS handles several at once.

SUMIF

Definition: Adds up the values in a range that meet exactly one condition.

Syntax: =SUMIF(range_to_check, condition, range_to_sum)

```
Dept column B2:B9, Salary column D2:D9
=SUMIF(B2:B9,"IT",D2:D9) -> 198000
```

SUMIFS

Definition: Adds up the values in a range that meet two or more conditions together. Note the sum range comes first here.

Syntax: =SUMIFS(range_to_sum, range1, condition1, range2, condition2, ...)

```
Dept=B2:B9, City=C2:C9, Salary=D2:D9
=SUMIFS(D2:D9,B2:B9,"HR",C2:C9,"Pune") -> 39000
```

Trainer tip: Keep every range the same size (same number of rows) in SUMIFS — mismatched ranges cause a #VALUE! error.

6. COUNTIF & COUNTIFS

Both functions count cells that meet a condition — *COUNTIF* for one condition, *COUNTIFS* for several at once.

COUNTIF

Definition: Counts how many cells in a range satisfy a single condition.

Syntax: =COUNTIF(range, condition)

```
=COUNTIF(B2:B9, "IT")    -> 3
=COUNTIF(D2:D9, ">50000")  -> 5
```

COUNTIFS

Definition: Counts how many rows satisfy two or more conditions together across different columns.

Syntax: =COUNTIFS(range1, condition1, range2, condition2, ...)

```
=COUNTIFS(B2:B9, "IT", C2:C9, "Pune")  -> 2
```

Trainer tip: A wildcard works inside *COUNTIF* too — =COUNTIF(A2:A9, "*Kumar*") counts every cell containing that text anywhere.

7. AVERAGEIF & AVERAGEIFS

Both functions average values that meet a condition — *AVERAGEIF* for one condition, *AVERAGEIFS* for several at once.

AVERAGEIF

Definition: Calculates the mean of the values in a range that meet a single condition.

Syntax: =AVERAGEIF(range_to_check, condition, range_to_average)

```
=AVERAGEIF(B2:B9, "Finance", D2:D9)  -> 81500
```

AVERAGEIFS

Definition: Calculates the mean of values that meet two or more conditions together.

Syntax: =AVERAGEIFS(range_to_average, range1, condition1, range2, condition2, ...)

```
=AVERAGEIFS(D2:D9, B2:B9, "IT", C2:C9, "Pune")  -> 63000
```

Trainer tip: If every row fails the condition, both functions return #DIV/0! — wrap them in IFERROR for a cleaner report.

8. Text Functions

These functions clean and reshape text data — pulling out parts of a string, removing extra spaces, replacing text, or joining values together.

LEFT

Definition: Returns a chosen number of characters starting from the left side of a text string.

Syntax: =LEFT(text, num_characters)

```
=LEFT("INV-2024-001",3) -> "INV"
```

RIGHT

Definition: Returns a chosen number of characters starting from the right side of a text string.

Syntax: =RIGHT(text, num_characters)

```
=RIGHT("INV-2024-001",3) -> "001"
```

MID

Definition: Returns a chosen number of characters starting from any position in the middle of a text string.

Syntax: =MID(text, start_position, num_characters)

```
=MID("INV-2024-001",5,4) -> "2024"
```

TRIM

Definition: Removes extra spaces from text, keeping only single spaces between words.

Syntax: =TRIM(text)

```
=TRIM(" Ravi Kumar ") -> "Ravi Kumar"
```

SUBSTITUTE

Definition: Replaces one piece of text with another, wherever it appears in a string.

Syntax: =SUBSTITUTE(text, old_text, new_text)

```
=SUBSTITUTE("Sales-Jan","-","/") -> "Sales/Jan"
```

TEXT

Definition: Converts a number or date into text, displayed in a format you choose.

Syntax: =TEXT(value, format_code)

```
=TEXT(45000,"₹#,##0") -> "₹45,000"
```

TEXTJOIN

Definition: Combines several values into one string using a chosen separator, and can skip blank cells.

Syntax: =TEXTJOIN(delimiter, ignore_empty, range)

```
=TEXTJOIN(", ", TRUE, A2:A5) -> "Ravi, Priya, Kiran, Deepa"
```

FIND

Definition: Returns the starting position of one text inside another. Case-sensitive.

Syntax: =FIND(find_text, within_text)

```
=FIND("-", "INV-2024") -> 4
```

SEARCH

Definition: Same as FIND but not case-sensitive, and also supports wildcards (* and ?).

Syntax: =SEARCH(find_text, within_text)

```
=SEARCH("inv", "INV-2024") -> 1 (matches regardless of case)
```

Trainer tip: Always TRIM imported data before running VLOOKUP or an exact-match formula on it — invisible spaces are one of the most common causes of a #N/A that 'should' match.

9. Date & Time Functions

Excel stores every date as a serial number counting days from 1-Jan-1900, which is why dates can be added, subtracted, and compared like ordinary numbers.

TODAY

Definition: Returns today's date and updates automatically every time the file recalculates.

Syntax: =TODAY()

```
=TODAY() -> 26-Jul-2026 (updates daily)
```

NOW

Definition: Returns the current date and time together, also auto-updating.

Syntax: =NOW()

```
=NOW() -> 26-Jul-2026 14:30
```

YEAR / MONTH / DAY

Definition: Extract just the year, month, or day number from a date.

Syntax: =YEAR(date) =MONTH(date) =DAY(date)

Date of Joining: 19-Mar-2019

```
=YEAR(B2) -> 2019
```

```
=MONTH(B2) -> 3
```

DATEDIF

Definition: Calculates the difference between two dates in years, months, or days — commonly used for age or years of experience.

Syntax: =DATEDIF(start_date, end_date, unit) unit: "Y"=years, "M"=months, "D"=days

=DATEDIF("19-Mar-2019",TODAY(),"Y") -> 7 (years of service)

NETWORKDAYS

Definition: Counts the working days between two dates, automatically excluding weekends.

Syntax: =NETWORKDAYS(start_date, end_date, [holidays])

=NETWORKDAYS("01-Jun-2026","30-Jun-2026") -> 22 working days

WORKDAY

Definition: Returns a date that is a given number of working days before or after a start date.

Syntax: =WORKDAY(start_date, days, [holidays])

=WORKDAY("01-Jun-2026",10) -> 15-Jun-2026

EDATE

Definition: Shifts a date forward or backward by a chosen number of whole months.

Syntax: =EDATE(start_date, months)

=EDATE("15-Jan-2026",3) -> 15-Apr-2026

EOMONTH

Definition: Returns the last day of the month, a chosen number of months before or after a start date.

Syntax: =EOMONTH(start_date, months)

=EOMONTH("15-Jan-2026",0) -> 31-Jan-2026

Trainer tip: DATEDIF works correctly but does not appear in Excel's autocomplete list — you must type it from memory, spelling included.

10. Logical Functions

These functions test a condition and decide what result to show, or catch errors before they display on screen.

IF

Definition: Checks one condition and returns one value if it is TRUE, another if FALSE.

Syntax: =IF(logical_test, value_if_true, value_if_false)

```
=IF(A2>=60,"Pass","Fail")
Score 75 -> Pass
```

Nested IF

Definition: Places one IF inside another to handle more than two possible outcomes.

Syntax: =IF(cond1, result1, IF(cond2, result2, result3))

```
=IF(A2>=90,"A",IF(A2>=75,"B",IF(A2>=60,"C","D")))
Score 88 -> B
```

IFERROR

Definition: Catches any formula error and replaces it with a custom message instead of #N/A or #DIV/0!.

Syntax: =IFERROR(formula, value_if_error)

```
=IFERROR(VLOOKUP("Ravi",A2:B9,2,0),"Not Found")
```

CHOOSE

Definition: Returns one value from a list, selected by its position number.

Syntax: =CHOOSE(index_number, value1, value2, ...)

```
=CHOOSE(2,"Red","Green","Blue") -> Green
```

Trainer tip: If a nested IF goes beyond 3-4 levels, switch to IFS instead — it is far easier to read and debug later.

11. VLOOKUP

VLOOKUP

Definition: Searches for a value in the first column of a table and returns a value from a chosen column in the same row. VLOOKUP stands for Vertical Lookup. It searches for a value in the first column of a table and returns a value from any other column in the same row. Think of it like searching for a student by their roll number and fetching their marks.

Syntax: =VLOOKUP(lookup_value, table_array, col_index_num, range_lookup) — use 0 or FALSE for an exact match

How to use:

1. Identify the lookup value, the table range, and the column number (counted from the lookup column as 1) that holds the answer.
2. Type the formula and lock the table_array with \$ signs before copying it down.
3. Set range_lookup to 0 (exact match) unless you specifically need an approximate match.

Name	Emp ID	Designation	Salary
Ravi Kumar	E101	Analyst	55000
Priya Shah	E102	Manager	45000

=VLOOKUP("Priya Shah",A2:D9,3,0) -> Manager

Trainer tip: If VLOOKUP returns #N/A even though the value looks correct, check for extra spaces or a text/number mismatch in the lookup column.

12. HLOOKUP

HLOOKUP

Definition: HLOOKUP stands for Horizontal Lookup. It is exactly like VLOOKUP but works on data that is arranged horizontally (across rows instead of down columns). If your data has months across columns, HLOOKUP is the right choice.

Syntax: =HLOOKUP(lookup_value, table_array, row_index_num, range_lookup)

	Jan	Feb	Mar
Sales	45000	52000	48000

=HLOOKUP("Feb",A1:D2,2,0) -> 52000

Trainer tip: HLOOKUP is used far less than VLOOKUP because most Excel data runs down in columns, not across in rows.

13. XLOOKUP

XLOOKUP

Definition: XLOOKUP is the modern, upgraded version of VLOOKUP. It was introduced in Excel 365 and Excel 2021. It removes all the limitations of VLOOKUP and is much easier to use. If your students are using a modern version of Excel, teach XLOOKUP as the standard going forward.

Syntax: =XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])

=XLOOKUP("Priya Shah",A2:A9,C2:C9,"Not Found") -> 45000

=XLOOKUP("Zoya",A2:A9,C2:C9,"Not Found") -> Not Found

Trainer tip: XLOOKUP does not need the lookup column to be first in the table, unlike VLOOKUP — it also works right-to-left.

14. INDEX-MATCH

Combining these two functions replicates VLOOKUP but works in any direction and does not break if columns are inserted.

MATCH

Definition: Finds the position of a value within a range.

Syntax: =MATCH(lookup_value, lookup_array, 0)

=MATCH("E102",B2:B9,0) -> 2 (2nd position in the range)

INDEX

Definition: Returns the value at a given position inside a range.

Syntax: =INDEX(range, row_number)

=INDEX(C2:C9,2) -> value in the 2nd row of that range

INDEX + MATCH (combined)

Definition: Nesting MATCH inside INDEX looks up a value anywhere in a table, in either direction.

Syntax: =INDEX(return_column, MATCH(lookup_value, lookup_column, 0))

=INDEX(C2:C9,MATCH("E102",B2:B9,0)) -> HR

Trainer tip: INDEX-MATCH is generally faster than VLOOKUP on large datasets and keeps working even if columns are added or removed from the table.

15. Information Functions (IS* Functions)

These functions check what is inside a cell and return TRUE or FALSE — the backbone of data validation and error handling.

ISNUMBER

Definition: Checks whether a cell contains a number.

Syntax: =ISNUMBER(value)

```
=ISNUMBER(100)  -> TRUE  
=ISNUMBER("Excel")  -> FALSE
```

ISTEXT

Definition: Checks whether a cell contains text.

Syntax: =ISTEXT(value)

```
=ISTEXT("Excel")  -> TRUE
```

ISBLANK

Definition: Checks whether a cell is genuinely empty.

Syntax: =ISBLANK(cell)

```
=ISBLANK(A5)  -> TRUE if A5 is empty
```

ISERROR

Definition: Checks whether a cell contains any type of error.

Syntax: =ISERROR(value)

```
=ISERROR(1/0)  -> TRUE
```

ISNA

Definition: Checks specifically for the #N/A error, most often from a failed lookup.

Syntax: =ISNA(value)

```
=ISNA(VLOOKUP("Zoya",A2:B9,2,0))  -> TRUE if not found
```

Trainer tip: *ISERROR treats every error the same way; use ISNA if you specifically need to tell a lookup failure apart from other errors.*

16. Advanced Filter

Advanced Filter

Definition: Filters data using a separate criteria range typed on the worksheet, supporting AND/OR logic and the option to copy results elsewhere — more powerful than the simple AutoFilter dropdowns.

How to use:

1. Above your data, type a small criteria range using the exact same column headers as your table.
2. Conditions on the same row = AND logic; conditions on separate rows = OR logic.
3. Go to Data tab -> Advanced (Sort & Filter group).
4. Set List range to your data, Criteria range to the small range you built, and click OK.

Criteria range:

Product	Region	Qty
Mobile	North	>3

-> only rows where all three conditions are true are shown

Trainer tip: The criteria range headers must match the data table's headers exactly (spelling included), or Advanced Filter silently returns zero results.

17. Data Validation

Data Validation restricts what a user is allowed to type into a cell, reducing data-entry mistakes.

Basic Validation Rule

Definition: Restricts a cell to a whole number, decimal, date, or text length within a range you set.

How to use:

1. Select the cell(s), go to Data tab -> Data Validation.
2. Under Allow, choose Whole Number, Decimal, Date, or Text Length and set the limits.
3. Add an Input Message and Error Alert on the other two tabs of the same dialog.

Allow: Whole Number | Data: between 18 and 60

Typing 15 -> blocked with the Error Alert message

Dropdown List

Definition: Restricts a cell to a fixed set of choices shown as a dropdown.

How to use:

1. Data tab -> Data Validation -> Allow: List.
2. Type the options separated by commas, or point Source to a range/Named Range.

Allow: List | Source: IT, HR, Finance

-> cell shows a dropdown with only these three options

Dependent (Cascading) Dropdown

Definition: A second dropdown whose options change based on what was chosen in the first one.

How to use:

1. Create one Named Range per parent category (e.g. North, South).
2. Set the second cell's Source to =INDIRECT(first_cell).

Region cell = North -> City dropdown Source: =INDIRECT(\$E2)
-> shows only cities that belong to North

Trainer tip: A Named Range used in INDIRECT must not contain spaces — replace 'West Zone' with 'West_Zone', or the dependent dropdown shows a blank list.

18. Conditional Formatting

Conditional Formatting automatically changes a cell's appearance based on a rule, so patterns and exceptions stand out without manual formatting.

Built-in Rules

Definition: Ready-made rules such as Highlight Duplicates, Top/Bottom values, Color Scales, and Icon Sets.

How to use:

1. Select the range.
2. Home tab -> Conditional Formatting -> pick a built-in rule (e.g. Top 10 Items).
3. Choose the formatting style and click OK.

Rule: Top 10% | Applied to Salary column
-> highest-paid employees are highlighted automatically

Custom Formula Rule

Definition: A rule written as a formula, useful for anything a built-in rule cannot do — like formatting a whole row based on one column.

How to use:

1. Home tab -> Conditional Formatting -> New Rule -> 'Use a formula to determine which cells to format'.
2. Write a formula that returns TRUE for the cells to format, e.g. =B2>75000.
3. Click Format, choose the style, and click OK.

Formula: =B2>75000 applied to range A2:C9
-> the entire row is filled green wherever Salary > 75000

Trainer tip: When formatting a whole row from one column's condition, lock only the column (\$B2) and leave the row unlocked — that is what makes the rule apply correctly to every row.

9. Pivot Tables

A Pivot Table summarizes a large dataset in seconds by dragging fields into Rows, Columns, Values, and Filters, instead of writing formulas.

Creating a Pivot Table

Definition: A Pivot Table is not a formula — it is a powerful built-in tool in Excel. It allows you to summarize, group, and analyze thousands of rows of data in seconds without writing a single formula. Think of it as a magic summary machine: you drag and drop fields and Excel instantly calculates totals, counts, and averages by any category.

How to use:

1. Click inside your data, then Insert tab -> PivotTable -> New Worksheet.
2. Drag a category field (e.g. Department) into ROWS.
3. Drag a numeric field (e.g. Salary) into VALUES — Excel sums it by default.
4. Optionally drag another field into COLUMNS or FILTERS to slice the data further.

Rows = Department | Values = Sum of Salary
 IT -> 188000
 HR -> 125000
 Finance -> 163000

Refreshing & Grouping

Definition: Keeps the pivot table in sync with the source data, and rolls dates up into months, quarters, or years.

How to use:

1. After source data changes, right-click inside the pivot -> Refresh (or press Alt+F5).
2. Right-click a date field -> Group -> choose Months/Quarters/Years.
3. Right-click a value -> Value Field Settings to switch Sum to Average/Count/Max.

Grouped by Quarter: Q1 -> 3 orders, Q2 -> 5 orders

Trainer tip: If new rows are added below the original range, update the source first (PivotTable Analyze -> Change Data Source) before refreshing, or the new rows are silently excluded.

20. Charts & Visualization

Choosing the right chart type makes a comparison instantly clear — each type is suited to a different kind of question.

Column / Bar Chart

Definition: Compares values across categories using vertical (column) or horizontal (bar) bars.

Department vs Total Salary -> one bar per department

Line Chart

Definition: Shows a trend over a continuous period, such as months or years.

Monthly Revenue Jan-Dec -> one continuous line

Pie Chart

Definition: Shows how parts make up a whole — best kept to 5-6 slices or fewer.

Market share by product: A 40%, B 35%, C 25%

Scatter Chart

Definition: Shows the relationship (correlation) between two numeric variables.

Experience (years) vs Salary -> each point is one employee

Combo Chart

Definition: Mixes two chart types with a secondary axis when the value scales are very different.

How to use:

1. Select the data, Insert a base chart, right-click one series -> Change Series Chart Type, tick 'Secondary Axis' for that series.

Revenue as columns (primary axis) + Profit Margin % as a line (secondary axis)

Trainer tip: Avoid 3D chart styles for anything you need to measure accurately — the tilt distorts how tall or how big each slice/bar actually looks.

21. Data Protection

Protects a worksheet or workbook from unwanted changes. By default every cell is 'Locked', but that only takes effect once protection is turned on.

Protect Sheet

Definition: Stops users from editing cells, formatting, or inserting rows within one worksheet.

How to use:

1. Select the cells users SHOULD be allowed to edit, press Ctrl+1 -> Protection tab -> untick Locked.
2. Review tab -> Protect Sheet, set a password (optional), click OK.

Locked: Employee Name, Emp ID, Department

Unlocked: Salary Adjustment column only

-> users can type only in the unlocked column

Protect Workbook

Definition: Stops structural changes — adding, deleting, hiding, or renaming sheets — across the whole file.

How to use:

1. Review tab -> Protect Workbook, set a password, click OK.

-> the sheet tabs can no longer be added, deleted, or renamed

Trainer tip: Hidden (for formulas) only takes effect once the sheet is protected — ticking Hidden alone does nothing until Protect Sheet is also applied.

22. Power Query

Power Query

Definition: Excel's built-in tool to import, clean, and combine data through a recorded, repeatable set of steps, loading the result as a refreshable table.

How to use:

1. Data tab -> Get Data -> From File / From Text-CSV, and select the source.
2. In the Query Editor, fix data types, rename columns, and apply cleaning steps — each is recorded in Applied Steps.
3. To combine tables: Merge Queries (like a SQL JOIN) or Append Queries (like a SQL UNION).
4. Click Close & Load; to refresh later, Data tab -> Refresh All.

Source: Monthly_Sales_Jan.csv + Monthly_Sales_Feb.csv

Step 1: Import both as queries

Step 2: Append Queries -> Sales_Combined table

Step 3: Remove Duplicates on Order ID

Trainer tip: Every step is repeatable — if the source file updates, one click on Refresh replays the whole cleanup automatically.

23. VBA & Macros

A Macro replays a sequence of actions automatically — useful for any repetitive task like formatting a weekly report.

What is Macro?

Definition: A Macro is a recorded set of steps that Excel can replay automatically. Imagine you perform the same 10 steps every Monday morning — open a file, format some columns, apply a filter, bold the headers. A Macro lets you record those steps once and then run them with a single click.

A Macro is simply an automation shortcut. No coding knowledge is required to create a basic Macro using the Record feature.

How to use:

1. View tab -> Macros -> Record Macro, name it (no spaces), perform the actions, then Stop Recording.
2. Run it later from View tab -> Macros -> Run, or assign it to a button (Developer tab -> Insert -> Button).

Recorded: select A1:C1, bold, fill yellow -> replays identically every time

VBA Macro (hand-written)

Definition: VBA stands for Visual Basic for Applications. It is the programming language that runs behind all Macros in Excel. When you record a Macro, Excel automatically writes VBA code for you behind the scenes.

You do not need to learn VBA to use Macros for simple tasks. But understanding VBA allows you to edit, customize, and build powerful automation that goes far beyond what the recorder can do.

How to use:

1. Press Alt+F11 to open the VBA Editor, then Insert -> Module.
2. Type code between Sub NameHere() and End Sub, using Dim for variables and Range/Cells to reference cells.
3. Save the file as .xlsm (Excel Macro-Enabled Workbook) to keep the code.

```
Sub HighlightLowSalary()
    Dim i As Integer
    For i = 2 To 9
        If Cells(i, 3).Value < 40000 Then
            Cells(i, 3).Interior.Color = RGB(255, 0, 0)
        End If
    Next i
End Sub
```

Trainer tip: Recorded macros act only on the exact cells recorded (Absolute mode) unless Relative Recording is switched on beforehand from the Developer tab.